Bhattraju Vighnesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.58
Sri Chaitanya Jr. Kalasala , Madhapur , Hyderabad	Class XII (SSC)	2024	98.60%
Sri Chaitanya Techno School, Kompally , Medchal District	Class X (SSC)	2022	10.00%

SKILLS

Programming Languages: Verilog | C | CPP | MATLAB | LaTeX

Software & Tools: Xilinx Vivado | Icarus Verilog | Logisim | KiCad | LTSpice | Linux | Altium Designer | Arduino IDE | Mathematica | GitHub

RELEVANT COURSES

Mathematics: Differential Equations (A+), Vector Calculus (A+), Probability & Random Processes, Matrix Theory

Computer Architecture: Microprocessors & Computer Architecture, Digital Systems

Algorithms & Programming: Data Structures & Algorithms, Introduction to Programming, Machine Learning,DevOps Fundamentals, Numerical Methods for Device Modelling.

PROJECTS

Bluetooth HID Controller Framework

- Designed a Bluetooth HID **controller framework** on ESP32, implementing **BLE-based wireless communication**, **event-driven input handling**, and **custom HID report generation** for low-latency Response.
- Built a modular firmware architecture supporting configurable button mappings and analog joystick inputs, enabling **real-time input transmission**, cross-platform compatibility, and effective wireless integration.

8-bit Breadboard Computer *(Digital Logic, Computer Architecture)*

- Designed and assembled an **8-bit breadboard computer** from **discrete logic components**, integrating the **CPU datapath**, **ALU**, **registers**, **memory**, and **control logic** to execute machine-level instructions while applying fundamental concepts of computer architecture, processor design, and hardware system integration.

RISC-V 32-bit Processor on FPGA | *Verilog, FPGA, Computer Architecture*

- Designed and implemented a **32-bit RISC-V 5-stage pipelined processor on FPGA** using Verilog, integrating **hazard detection** and **data forwarding** to minimize pipeline stalls and improve instruction throughput, with functionality verified through simulation and hardware validation.

MiniGit - Version Control System

- Designed and implemented a Git-inspired version control system, building **content-addressable object storage** (Blobs, Trees, Commits), **hash-based indexing**, **DAG-based commit graph**, and **persistent object storage** for efficient repository management, enabling reliable version tracking and history traversal across multiple Branches.
- Developed version control operations including **repository initialization**, **branching**, **checkout**, **commit**, **merge**, and **object serialization**, while implementing an **interactive commit-tree visualizer** to graphically represent repository history.

FPGA-Based Real-Time Image Processing Pipeline | *Verilog, FPGA, AXI, Digital Design*

- Designed and implemented a **real-time FPGA-based image processing pipeline** for **2D convolution-based image blurring**, leveraging **line buffers** and **AXI streaming interfaces** to enable low-latency, high-throughput pixel processing.

POSITIONS OF RESPONSIBILITY

Transport & Logistics Coordinator, Elan & nVision

- Coordinated end-to-end transportation logistics for **IIT Hyderabad's annual techno-cultural festival** with a **10,000+ attendee** footfall, ensuring timely movement of artists, judges, VIP guests, and event personnel.
- **Planned** and managed travel itineraries, airport/railway pickups, local transportation, accommodation coordination, and **schedule adherence** while collaborating with hospitality, event management, and external transport vendors.
- Handled **real-time operational challenges** by coordinating across multiple teams, optimizing vehicle allocation and travel schedules to ensure smooth execution of performances, competitions, and guest engagements throughout the fest.

Class Representative, IC Design & Technology Branch

- Led coordination between students and faculty for **academic planning**, **examination scheduling**, and **administrative operations**, facilitating student requests across multiple courses, ensuring timely issue resolution.
- Initiated and organized **official batch networking events**, coordinating **l**ogistics, schedules, and stakeholders across all the batches to strengthen collaboration and **mentorship** within the department.

EXTRACURRICULAR

National Cadet Corps (NCC) Cadet

Achieved **B Certificate** and selected to participate in CAT-VII (Combined Annual Training Camp), gaining experience in leadership, discipline, teamwork, and outdoor training .